

Series and infinite products – 4

Finding the sum: telescoping, Abel, generating functions

Homework

12 сентября 2026 года

Theoretical minimum. Four devices cover almost everything. Telescoping after partial fractions, or after the addition formula for arctan, or after a trigonometric identity that halves the argument. Summation by parts, which removes a partial sum such as H_n from under the summation sign. Generating functions: a linear recurrence with polynomial coefficients turns $\sum a_n x^n$ or $\sum a_n x^n / n!$ into a differential equation. And the integral representation $\frac{1}{n} = \int_0^1 x^{n-1} dx$, which converts a series into an integral of a geometric progression. Expect $\ln 2$, $\zeta(2)$, $\zeta(3)$ and γ in the answers; the appearance of γ always means that the second term of the asymptotics of H_n has been used.

1. Using Abel summation compute $\sum_{n \geq 1} n x^n$, $\sum_{n \geq 1} n^2 x^n$ and $\sum_{n \geq 1} \frac{H_n}{2^n}$ for $|x| < 1$, and state in each case why the rearrangement is legitimate.
2. (OMOUS 2021) Define the sequence $(a_n)_{n \geq 0}$ by $a_0 = a_1 = 1$ and $a_n = a_{n-1} + 2(n-1)a_{n-2}$ for $n \geq 2$. Find the sum of the series $\sum_{n=0}^{\infty} \frac{a_n}{n!}$.
3. (Gazeta Matematica) Prove that $\sum_{k=1}^{\infty} k(k+1-\zeta(2)-\zeta(3)-\dots-\zeta(k+1)) = \zeta(3)$.
4. (Gazeta Matematica, problem 436) Calculate $\sum_{n=1}^{\infty} \left(1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln \sqrt{n(n+1)} - \gamma\right)$, where γ is the Euler–Mascheroni constant.
5. (IMC) For each $n \geq 1$ let $a_n = \sum_{k=0}^{\infty} \frac{k^n}{k!}$ and $b_n = \sum_{k=0}^{\infty} (-1)^k \frac{k^n}{k!}$. Show that $a_n b_n$ is an integer.

6. Prove that $\sum_{n \geq 1} \frac{H_n}{n(n+1)} = \zeta(2)$ and $\sum_{n \geq 1} \frac{H_n}{n^2} = 2\zeta(3)$.
7. Prove that $\sum_{n \geq 1} \frac{(-1)^{n-1} H_n}{n} = \frac{\zeta(2)}{2} - \frac{\ln^2 2}{2}$.
8. For $0 < a < b$ compute $\sum_{n \geq 0} \frac{1}{(n+a)(n+b)}$ through the integral representation, and deduce $\sum_{n \geq 1} \frac{1}{n(n+a)}$ in terms of the logarithmic derivative of the gamma function.
9. (ISTCiM 2024) Calculate the sum $\sum_{n=1}^{\infty} (-1)^{n-1} \left(n \left(\sum_{k=n+1}^{\infty} \frac{1}{k^2} \right) - 1 \right)$.
10. Prove that $\sum_{n \geq 2} (\zeta(n) - 1) = 1$ and $\sum_{n \geq 2} \frac{\zeta(n)-1}{n} = 1 - \gamma$, and say exactly which interchange of the order of summation is used in each case (the general rule is in Series and infinite products – 5).